VERILOG ROADMAP

Complete Curriculum from Basics to Advanced RTL & Verification

LEVEL 1 — ABSOLUTE BASICS

01	AND gate	**11**	1:4 Demultiplexer
02	OR gate	**12**	Buffer
03	NOT gate	**13**	Tri-state buffer
04	NAND gate	**14**	Basic comparator
05	NOR gate	**15**	2-bit comparator
06	XOR gate	**16**	Basic gates using assign
07	XNOR gate	**17**	Wire and Reg usage
08	2:1 Multiplexer	**18**	Simple combinational circuit
09	4:1 Multiplexer	**19**	Operator practice
10	1:2 Demultiplexer	**20**	Bitwise operations

LEVEL 2 — COMBINATIONAL CIRCUITS

21	Half Adder	**31**	Gray-to-Binary converter
22	Full Adder	**32**	Binary-to-BCD converter
23	Half Subtractor	**33**	BCD-to-Binary converter
24	Full Subtractor	**34**	BCD-to-7-Segment decoder
25	4-bit Ripple Carry Adder	**35**	3-to-8 Decoder
26	4-bit Adder/Subtractor	**36**	8-to-3 Encoder
27	2-bit ALU	**37**	4-to-2 Priority Encoder
28	4-bit ALU	**38**	8-to-3 Priority Encoder
29	8-bit ALU	**39**	16-to-4 Priority Encoder
30	Binary-to-Gray converter	**40**	Parity Generator

LEVEL 3 — SEQUENTIAL LOGIC

41	Parity Checker	**51**	4-bit Register
42	D Flip-Flop	**52**	8-bit Register
43	D Flip-Flop with Reset	**53**	Shift Register
44	D Flip-Flop with Set	**54**	SISO [Serial Input, Serial Output]
45	D Flip-Flop with Set and Reset	**55**	SIPO [Serial Input, Parallel Output]
46	D Flip-Flop with Enable	**56**	PISO [Parallel Input, Serial Output]
47	T Flip-Flop	**57**	PIPO [Parallel Input, Parallel Output]
48	JK Flip-Flop	**58**	Universal Shift Register
49	SR Flip-Flop	**59**	Parallel Load Register
50	Register	**60**	Register with Enable

LEVEL 4 — COUNTERS

61	Up Counter	**71**	Decade Counter
62	Down Counter	**72**	Ring Counter
63	Up/Down Counter	**73**	Johnson Counter
64	2-bit Counter	**74**	Gray Counter
65	4-bit Counter	**75**	Programmable Counter
66	Parameterized N-bit Counter	**76**	Counter with Enable
67	Mod-2 Counter	**77**	Counter with Load
68	Mod-3 Counter	**78**	Counter with Terminal Count
69	Mod-5 Counter	**79**	Frequency Divider
70	Mod-10 Counter	**80**	Clock Divider

LEVEL 5 — FSM (FINITE STATE MACHINES)

81	Moore FSM — sequence detector	**91**	Pedestrian Traffic Controller
82	Mealy FSM — sequence detector	**92**	UART Transmitter FSM
83	101 sequence detector — overlapping	**93**	UART Receiver FSM
84	101 sequence detector — non-overlapping	**94**	Arbiter FSM
85	1101 detector	**95**	FSM with timeout
86	Traffic Light Controller	**96**	FSM with error state
87	Vending Machine	**97**	FSM-based counter
88	Elevator Controller	**98**	FSM-based protocol controller
89	Washing Machine Controller	**99**	FSM with multiple outputs
90	Digital Lock	**100**	Parameterized FSM

LEVEL 6 — MEMORIES

101 ROM	**111** Memory with word enable
102 8x8 RAM	**112** Memory Initialization
103 16x8 RAM	**113** Read-First RAM
104 Single-port RAM	**114** Write-First RAM
105 Dual-port RAM	**115** No Change RAM
106 Synchronous RAM	**116** Parameterized RAM
107 Asynchronous RAM	**117** RAM with parity
108 Register File	**118** ROM with case
109 2-read/1-write Register File	**119** ROM with lookup table
110 Memory with byte enable	**120** Small memory-based design

LEVEL 7 — PRACTICAL RTL DESIGNS

121 UART Transmitter	**131** Pulse Counter
122 UART Receiver	**132** Edge Detector
123 Complete UART	**133** Rising Edge Detector
124 SPI Master	**134** Falling Edge Detector
125 SPI Slave	**135** Both-Edge Detector
126 I2C Master simplified	**136** Clock Enable Generator
127 I2C Slave simplified	**137** Debouncer
128 PWM Generator	**138** Frequency Counter
129 PWM with Duty Control	**139** Digital Stopwatch
130 Pulse Generator	**140** Digital Timer

LEVEL 8 — FIFOS & ADVANCED RTL

141 Synchronous FIFO	**151** Valid-Ready Interface
142 FIFO with Full/Empty flags	**152** Ready-Valid FIFO
143 Parameterized FIFO	**153** Round-Robin Arbiter
144 Asynchronous FIFO	**154** Fixed-Priority Arbiter
145 Gray-code Async FIFO	**155** Priority Arbiter with Mask
146 FIFO with Almost-Full	**156** Crossbar Switch
147 FIFO with Almost-Empty	**157** Packet Counter
148 Skid Buffer	**158** Packet Parser
149 Elastic Buffer	**159** Stream Data Processor
150 Pipeline Register	**160** Simple DMA Controller

LEVEL 9 — CPU / RTL ARCHITECTURE

161 Single-cycle ALU	**171** RISC-V Register File
162 Register File	**172** RISC-V Immediate Generator
163 Program Counter	**173** RISC-V Control Unit
164 Instruction Memory	**174** RISC-V Single-Cycle CPU
165 Data Memory	**175** RISC-V Branch Unit
166 Instruction Decoder	**176** RISC-V Load/Store Unit
167 Control Unit	**177** RISC-V Pipeline — basic
168 Simple 8-bit CPU	**178** 2-stage Pipeline
169 Simple RISC CPU	**179** 3-stage Pipeline
170 RISC-V ALU	**180** 5-stage RISC-V pipeline

LEVEL 10 — ADVANCED VLSI RTL

181 Clock Gating	**191** APB Slave
182 Power Gating Control	**192** APB Register Bank
183 Reset Synchronizer	**193** Simple AXI-Lite Slave
184 Clock-Domain Crossing Sync	**194** Watchdog Timer
185 Pulse Synchronizer	**195** Interrupt Controller
186 Handshake CDC	**196** Timer Peripheral
187 Asynchronous FIFO CDC	**197** GPIO Controller
188 Multi-cycle Pipeline	**198** SPI Peripheral with Reg IF
189 Configurable Pipeline	**199** UART Peripheral with Reg IF
190 AXI-Stream-like Interface	**200** SoC Peripheral Bus

LEVEL 11 — VERIFICATION ORIENTED

201 Self-checking testbench	**211** FIFO verification environment
202 Testbench using tasks	**212** UART verification environment
203 Testbench using functions	**213** ALU verification environment
204 Random stimulus generator	**214** FSM verification environment
205 Directed vs random testbench	**215** RAM verification environment
206 Scoreboard	**216** Coverage Group
207 Reference model	**217** Assertion-based verification
208 Monitor	**218** Constrained Random TB
209 Functional coverage	**219** Functional Coverage in TB
210 Assertions	**220** Complete verification flow

LEVEL 12 — SYSTEMVERILOG / UVM

206 SystemVerilog interface	**214** Environment
207 SystemVerilog clocking block	**215** SystemVerilog assertions
208 Transaction class	**216** Functional coverage
209 Generator	**217** Complete UVM ALU verification
210 Driver	**218** Complete UVM FIFO verification
211 Monitor	**219** Complete UVM UART verification
212 Scoreboard	**220** Complete UVM APB verification
213 Agent	

Course Curriculum & Roadmap Credits:
The sequential structure and syllabus of these 220 practice modules are based on the **Verilog Roadmap** designed and compiled by **Suyash Ghadage**. All rights and core academic credits for this learning path belong exclusively to the original creator.
Source Reference: ***Suyash Ghadage - Verilog Roadmap on LinkedIn***